MILITARY SPECIFICATION

PROJECTILE, 81MM, ILLUMINATING, M301A3, PARTS LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-60447A(PA), dated 15 March 1971, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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2.1: Military Specifications: Add the following:

"MIL-P-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for"

Military Standard: Add the following:

"MIL-STD-1169 - Packaging Packing and Marking for Shipment of Inert Ammunition Components"

2.1: Under Publications delete:

"ET-8865054 - Equipment Tabulations"

Add new Inspection Equipment Drawings:

"INSPECTION EQUIPMENT DRAWINGS

9201136	-	Tunnel, Light
9201268	-	Procedure, Light Output Measurement
9201390	-	Procedure, Photocell Checkout
9247071	-	Photocell
9200964	-	Photometer Head with Cell
7297287	-	Chamber Gage
7257965	-	Limit Check Gage

FSC 1315

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2.2: Delete paragraph 2.2 in its entirety and substitute the following new paragraph:

"2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

TRANSPORTATION

49-CFR 100-199 - Code of Federal Regulations for the Transportation of Explosives and other Dangerous Articles.

The code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "CFR-100-199 (latest revision)".

AMERICAN SOCIETY FOR TESTING MATERIAL (ASTM)

A-513 - Specification for Electric-Welded Steel Tubing
E-18 - Rockwell Hardness Testing
E-8 - Tension Testing of Metallic Materials."

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3.7: Delete paragraph in its entirety and substitute the following:

"3.7 Static test of case loading assembly.

3.7.1 Burning time. The burning time shall not be less than 65 seconds (sec).

3.7.2 Luminosity. The average luminosity shall not be less than 336,000 candlepower minimum (min.)."

3.8.1, 2nd sentence, Proving Ground: Change to read:

"(The requirements of 3.8.2, 3.8.3, 4.2.1e, 4.2.1f, 4.2.3 and 4.2.4 are not applicable ...)"

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* 3.11: Delete paragraph in its entirety.

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4.2.1a: Change to read "ten" in place of "fifteen".

4.2.1c: Delete in its entirety.

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4.3.2.1, Cone Tail (Prior to Paint) (See Dwg. 9232482):

Delete Major 101 in its entirety and substitute:

"101 True position of pitch diameter of fin threads Gage".

Delete Major defect 102 in its entirety.

Delete Major defects 108 and 109 and substitute the following:

"108 Run out of rear outside diameter of cone Gage 01008
109 Run out of outside taper Gage 01009".

Add the following new Major defect:

"110 Roundness of forward outside diameter Gage 01021".

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4.3.2.2, Cone Tail (After Paint): Delete Major defect 104 and substitute the following:

"104 Runout of bourrelet Gage 02012".

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4.3.2.4, Adapter (See Dwg. 9232483):

Delete Major defect 101 in its entirety and substitute:

"101 True position of pitch diameter threads (Tolerance zone projected 3.3") Gage".

Delete Major defect 106 in its entirety and substitute:

"106 True position of pitch diameter threads (Tolerance zone projected 0.25") Gage".

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Add the following new Major defects:

107 Perpendicularity of diameter at rear shoulder with rear shoulder (datum B) Gage 04015
108 Runout of front chamfer face Gage 04016".

Change Minor AQL from "2.5" to "4.0".

Change Minor defect 201 to read:

"201 Length of thread including thread relief Gage 04006".

Add the following new Minor defect:

"208 Max thread relief Gage 04013".

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4.3.2.5, Tube Body (See Dwg. 9232484):

Delete Major defects 109, 110, and 111 and substitute the following new Major defects 109, 110 and 111:

"109 Runout of outside diameter Gage 05009
110 Runout of bourrelet Gage 05010
111 Runout of adapter cavity Gage 05011".

Add the following new Major defect:

"112 Perpendicularity of end face with tail cone cavity Gage 05022".

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4.3.2.6, Metal Parts Assembly (Prior to Painting) (See Dwg. 9232481):

Delete Major defect 101 in its entirety and substitute:

"101 True position of outside diameter of tube Gage".

Delete Major defect 106 in its entirety and substitute:

"106 True position of pitch diameter Gage".

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4.3.2.14, Illuminant Assembly (See Dwg. 8865056): Add the following to Major 101:

"including pressure tape and twist tie wire when applicable".

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4.3.2.16, Loading Assembling, After Assembly of Shear Pins:

Add new special inspection category:

"SPECIAL: Depth and diameter of shear pin holes (4) gage (see paragraph 4.3.4.3)."

* Major and minor defects only, delete in their entirety and substitute the following:

"Major:	AQL 0.25 percent		Code No.
101	Army shear pin above flush	Visual	16003
102	Depth of shear pin (max.)	Gage	16004
Major:	AQL 0.65 percent		
103	Tail cone improperly assembled to body	Visual	16005
104	Location of shear pins	Gage	16006
105	Excessive gap between tail cone and body tube	Gage	16007
106	Presence of cracks or acute discontinuities	Visual	16012
Minor:	AQL 1.5 percent		
201	Protective dust plug missing or loose	Visual/Manual	16010
202	Marking misleading or unidentifiable	Visual	16010
203	Evidence of poor workmanship	Visual	16011"

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4.3.2.18, Parachute Assembly, Prior to Inserting in Parachute Holder (Dwg. 9242684): Add the following new Major defect:

"104 Presence of less than 3 rows of stitching of skirt tape to shroud line Visual 18010".

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Add new paragraph:

"4.3.2.19 Groove-pin (see dwg 8882169 covering a detail of dwg 8865054)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.50 percent		
101	Expanded diameter	Gage	19001
102	Length	Gage	19002
Minor:	AQL 1.5 percent		
201	Diameter	Gage	19003
202	Surface finish, improper	Visual (see 6.3)	19004
203	Evidence of poor workmanship (see 3.9)	Visual (see 6.2)	

4.3.3.1, Static Functioning: Delete paragraph in its entirety and substitute the following:

"4.3.3.1 Static functioning (see 3.7). Major defect - Code No. 14008. The following sampling plan will apply with the sample tested as specified in 4.4.1.

4.3.3.1.1 Each inspection lot will consist of approximately eighty-five candles produced consecutively by the mixing/pressing operation and will be designated as one batch (see 6.4) for identification purposes.

4.3.3.1.1.1 Select a sample of three candles from each inspection lot (approximately 85 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.7. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements.

4.3.3.1.1.2 After successful completion of 4.3.3.1.1.1, the sampling frequency may be reduced to three candles from each group of six inspection lots (approximately 510 candles). The canister assembly shall be observed for compliance with the

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requirements specified in 3.7. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.3.3.1.1.1.

4.3.3.1.1.3 After successful completion of 4.3.3.1.1.2, the sampling frequency may be reduced to three candles from each group of inspection lots (approximately 1275 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.7. Continue this sampling frequency until 15 consecutive samples (15 groups of 3 candles) comply with the specified requirements. In the event the one lot fails to comply with the requirements, the sampling shall revert to 4.3.3.1.1.2.

4.3.3.1.1.4 After successful completion of 4.3.3.1.1.3, the sampling frequency may be reduced to three candles from each eight hours of production or approximately 1615 candles, whichever is less. The canister assembly shall be observed for compliance with the requirements specified in 3.7. In the event that one lot fails to comply with the requirements, the sampling shall revert to 4.3.3.1.1.3.

4.3.3.1.1.5 In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.3.3.1.1.2 and in the event of production stoppage greater than two weeks, the sampling plan shall revert to 4.3.3.1.1.1. If the continuous mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall revert to 4.3.3.1.1.2.

4.3.3.1.1.6 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot."

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4.3.3.3, Pull Test of Illuminant Assembly: Delete paragraph in its entirety and subtitute the following new paragraph:

"4.3.3.3 Pull Test of Illuminant Assembly (See Dwg. 8865056). Major Defect - Code No. 14011. The oval sleeve, suspension wire assembly and suspension cup shall be tested 100% prior to assembly and attachment to illuminant case with 6 rivets. Any assembly that visibly yields under the load shall be removed from the lot."

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4.3.3.4, Non-destructive Electrical Test of Body Tubes: At the end of last sentence delete words "paragraph S7".

4.3.3.6: Delete paragraph in it entirety and substitute the following new paragraph:

"4.3.3.6 Profile and alignment (See Dwg. 8865054) - Major defect - Code No. 16013. The loading assembly shall be gaged 100 percent for profile and alignment. Any loading assembly which fails to comply with the requirement specified shall be classed defective and removed from the lot. This requirement is not applicable when the contractor for the loading assembly is also the contractor for the complete round (see 6.1d)."

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4.3.3.7.1.3, Adapter (See Dwg. 9232483): Delete paragraph in its entirety and substitute the following new paragraph:

"4.3.3.7.1.3 Adapter (See Dwg. 9232483) - Major defect - Code No. 04013. After final heat treatment or stress relief a kiln block representing one lot of adapters shall be submitted so that two round specimens can be machined with minimum diameter - "D - 0.250" per fig. 8 ASTM - E8 (Federal Test Method Std. -151B) and the specimens shall be subjected to the tension test. Failure to comply with the requirements of the applicable drawings and specification shall be cause for rejection of the lot."

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* 4.3.3.9, Push Test of Shear Pins: Delete paragraph in its entirety.

4.3.3.10, Test of Shroud Lines: Delete paragraph in its entirety and substitute the following:

"4.3.3.10 Test of Shroud Lines (See Dwg. 9242684) - Major defect - Code No. 18011. As an in-process control, each machine sewing shroud lines shall sew one shroud line across a test canopy every two hours during production. The sewn shroud line shall be subjected to a load of 99-100 lbs across the canopy. If the thread holding the shroud line breaks, the product produced during the two hour interval on that machine shall be rejected. Then the thread, tape, and feed tensions shall be adjusted so that the shroud line can be loaded and comply with the test requirement. Ten lines shall pass the load test after adjustment prior to going back into production on the machine."

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4.3.4, Inspection Equipment: Delete paragraphs 4.3.4, 4.3.4.1 and 4.3.4.2 in their entirety and substitute the following new paragraph:

"4.3.4 Inspection Equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in "Method of Inspection" starting with paragraphs 4.3.2.1 and under paragraph 4.4 "Test Methods and Procedures". The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078."

Add new paragraphs as follows:

"4.3.4.3 The depth and diameter of the shear pin holes shall be inspected in the following manner using approved gages."

"4.3.4.3.1 Beginning with the first production shift, ten (10) consecutive drilled units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every 1/2 hour until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the machine set up shall be adjusted until ten (10) consecutive units are defect free."

"4.3.4.3.2 After succesfully completing the sampling of paragraph 4.3.4.3.1 for one eight hour production shift with no defects, the sampling plan may be reduced to five (5) drilled units being inspected every one hour and at the beginning and end of the shift. If a defect is found the sampling plan of paragraph 4.3.4.3.1 will be reinstated."

"4.3.4.3.3 After successfully completing the sampling plan of paragraph 4.3.4.3.2 for one eight hour production shift with no defects the sampling plan may be reduced to three (3) drilled units being inspected every hour and at the beginning and end of the shift. If a defect is found the sampling plan of paragraph 4.3.4.3.2 will be reinstated."

"4.3.4.3.4 After successfully completing the sampling plan of paragraph 4.3.4.3.3 for one eight hour shift with no defects, the sampling plan may be reduced to two (2) drilled unit being inspected every two hours. If a defect is found the sampling plan of paragraph 4.3.4.3.3 will be reinstated."

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"4.3.4.3.5 A change of tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change."

"4.3.4.3.6 A stop in production of over two months will require starting the inspection at the first production shift plan."

* "4.3.4.3.7: Delete paragraph in its entirety.

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4.4.1, Static Functioning Test: Delete paragraph in its entirety and substitute the following new paragraph:

"4.4.1 Static test of illuminant canister assembly. The equipment required for this test is delineated in 2.1 under Inspection Equipment Drawings. The temperature of the candle shall be $70^{o} \pm 5^{o}F$ at the time of test. A sample of three candles shall be selected to represent the mix/batch of candles (see 6.4). In calculating the candlepower of any one illuminant, the best continuous sixty seconds of illuminant readings are to be used. The first fire disc assemblies should be removed prior to testing. The distance from the photometric cell to the flare shall be $60 \pm .25$ feet. The axial center of the fireball shall be located $54 \pm .5$ inch from the floor. The temporary back wall listed on Dwg. 9201136 shall be removed so that the distance from the center of the candle to the back wall is from 14 to 15 feet. The center line of the baffle opening shall be in line with the axial center of the fire ball and the axial center of the photometric cell. The test shall be conducted in a hearth that has been painted flat black. Depending on the ash build-up, the tunnel hearth shall be repainted at least once in a three month duration to maintain the black interior, or more frequently as determined by the Government Quality Assurance Representative. If the burning time of an individual sample is less than 65 secs., or the average luminosity of less than 336,000 candlepower, the mix/batch shall be rejected.

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4.4.6.3, Adapter: Add the following new paragraph:

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"4.4.6.3 Adapter. The two specimens selected in accordance with 4.3.3.7.1.3 shall be tested for yield strength and elongation by either the offset method or the extension under load method as prescribed by ASTM-E8. The limiting set shall be 0.1 percent (0.001 inches per inch of gage length)."

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* **4.4.9:** Delete paragraph in its entirety.

4.4.10: Delete paragraph in its entirety

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6.1, Ordering Data: Delete paragraph 6.1 in its entirety and substitute:

"d. True Position. True Position requirements for Tube, Bodies; Body Tube Assemblies; cone, tails and tail cone assemblies with the exception of Runout defects "01008 and 01009" and True Position defect Code "06001", may be liberalized or omitted by mutual written agreement between the Government and Prime Contractor for the projectile loading assembly".

6.4: Delete paragraph in its and entirety and substitute the following new paragraph:

"6.4 Mix/Batch. Illuminant from the mixer will be collected in containers of approximately equal quantity of illuminant per container which will be numbered and used in sequence. Case loading assemblies will be assigned an inspection lot number for each 85 candles produced.

6.5, Inspection Equipment: Delete paragraphs 6.5, 6.5.1, 6.5.2,, 6.5.3, 6.5.4 and 6.5.5 in their entirety and substitute the following new paragraph:

"6.5 Submission of Designs for Approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAT-I(D), Dover, New Jersey 07801-5001".

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